An Evaluation of User Perception and Interaction in Immersive VR Restaurant Menus

Courtney Malone
Colorado State University
Fort Collins, Colorado, USA
courtney.malone@colostate.edu

Enock Apedjinou
Colorado State University
Fort Collins, Colorado, USA
enock.apedjinou@colostate.edu

ABSTRACT

This paper introduces an immersive VR restaurant menu system featuring interactive 3D food visualization and multimodal hand-tracking interaction.To gain deeper insight into user perceptions, participants completed structured interaction sessions followed by semi-structured interviews focused on usability, visual quality, comfort, and decision confidence. Participant responses were analyzed to identify patterns in user experience and evaluate perceptions of VR-based restaurant menu interfaces.

CCS Concepts

• **Human-centered computing → Virtual Reality**; **Interaction techniques**.

KEYWORDS

VR, Human Computer Interaction, Virtual Reality Menus, Multimodal Interaction

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1 INTRODUCTION

The restaurant experience is fundamentally mediated by information: menus shape not only what diners choose, but how they understand and evaluate food. Traditional menus, whether physical or digital, rely heavily on abstract text and static imagery, which often fail to convey the spatial, sensory, and experiential qualities of dishes. As virtual reality (VR) becomes increasingly capable of supporting rich, embodied interaction, it opens new possibilities for designing more immersive and intuitive dining interfaces.

Prior research in VR interaction has explored object manipulation, hand tracking, and multimodal input as ways to enhance user engagement and decision-making in virtual environments. These techniques have been shown to improve presence and interaction quality, particularly in contexts where spatial understanding is important. However, there remains a notable gap in VR system interaction tasks [5].

To address this gap, we present a VR restaurant menu system designed to explore how spatially embodied interfaces can support food selection and decision-making. This paper investigates how immersive representations of menu items may influence user understanding, preference formation, and interaction efficiency in a restaurant context.

We present a virtual reality restaurant menu system that enables immersive food selection through interactive 3D food models, hand tracking, and a virtual cart system. The system explores how multimodal interaction can improve user understanding and engagement in digital restaurant environments. Based on this motivation and prior work, we investigate the following research question: How does interaction with 3D food models in a VR restaurant environment affect user perception of usability, visualization quality, and decision confidence?

Authors' Contact Information: Courtney Malone, Colorado State University, Fort Collins, Colorado, USA, courtney.malone@colostate.edu; Enock Apedjinou, Colorado State University, Fort Collins, Colorado, USA, enock.apedjinou@colostate.edu.

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2 RELATED WORK

VR and AR in Food and Dining Contexts. Researchers have increasingly explored how immersive technologies can reshape food-related experiences. Attar et al. [2] used virtual reality food environments to study individual food consumer behavior in urban settings, demonstrating that VR can replicate ecologically valid decision-making contexts without requiring physical food environments. Complementing this, Nordbo et al. [10] developed Virtual Food Court, a VR environment specifically designed to assess how people make food choices under realistic simulated conditions, establishing early empirical grounding for VR as a food behavior research tool. On the augmented reality side, Alva et al. [1] proposed an AR visualization system for restaurant menu items using image reconstruction, showing that overlaying 3D representations of dishes onto physical menus improves diner understanding of food presentation. Marques et al. [8] extended the application of VR in food contexts further by exploring its therapeutic potential, using virtual food exposure environments to treat sensory food aversion an application that underscores the behavioral power of immersive food visualization beyond entertainment.

Recent work has also explored gaze-based interaction techniques as alternatives to traditional controller-driven navigation in immersive environments. [7] investigated gaze-based menu navigation in VR using both pie and list menu layouts, comparing interaction mechanisms including dwell selection, border-crossing, controller input, and multimodal gaze-plus-button interaction. Their findings highlight the importance of interaction modality and menu structure in shaping usability and navigation efficiency in XR systems. Prior studies referenced in their work further suggest that users often prefer list-based menu layouts over radial or pie menus in VR contexts. These findings are relevant to our work as our VR restaurant prototype similarly explores multimodal interaction approaches designed to improve usability, reduce interaction difficulty, and support accessible menu navigation during food selection tasks.

Importantly, prior studies have also evaluated the validity of VR-based consumer research by comparing behavior in virtual environments with real-life (RL) settings. These comparisons suggest that VR can produce behaviorally comparable outcomes to RL conditions under certain experimental designs, reinforcing its potential as a methodologically robust research tool for food-related studies.

Multisensory and Multimodal Human-Food Interaction.A substantial body of work has examined how engaging multiple senses simultaneously shapes food perception and experience. Velasco et al. [12] organized and synthesized the growing research community around multisensory approaches to human-food interaction, identifying technologies, mechanisms, and open challenges across engineering, psychology, and food science. Narumi [9] advanced this space through multi-sensorial VR and augmented human-food interaction, demonstrating systems such as Meta-Cookie, a flavor augmentation interface that modifies perceived taste through overlaid visual and olfactory information and Augmented Satiety, which manipulates food intake perception through AR. These works establish that the visual representation of food in immersive environments is not merely aesthetic but can directly affect perception and behavior, motivating our focus on high-quality 3D food model visualization as a core design goal.

AI, Personalization, and Digital Ordering Interfaces. As digital food interfaces grow more sophisticated, researchers have explored personalization and intelligent assistance as means of improving the ordering experience. Wong et al. [15] examined the effects of customizing an AI shopping assistant in a VR retail environment, finding that tailored assistant behavior increased user engagement and purchase confidence, findings that transfer directly to VR menu contexts. Vijjali and Bhageria [13] developed FoodNet, a system for simplifying online food ordering through contextual food combinations, demonstrating that intelligent suggestion systems can meaningfully reduce decision load during digital ordering. Wacker [14] evaluated menu techniques for handheld AR interfaces combining a smartphone and mid-air pen, contributing empirical insight into how input modality affects menu navigation efficiency directly relevant to our comparison of hand tracking and voice command interaction in a headset-based context.

Accessibility and Inclusive Design in VR. As VR systems are deployed to broader user populations, accessibility becomes a central design concern. Forssell et al. [6] investigated inclusive VR gaming with older adults, assessing the accessibility of entertainment and educational VR applications and surfacing design considerations around ease of interaction, physical comfort, and learnability for users with limited prior VR experience. Their findings are directly relevant to our study, in which participants with no prior VR experience reported greater difficulty and discomfort than experienced users, underscoring the need for accessible onboarding and interaction design in consumer-facing VR applications.

Despite these contributions across immersive food environments, multisensory interaction, digital ordering, and VR accessibility, no prior work has directly examined how hand-tracked, controller-free VR interaction shapes food selection behavior and decision confidence in a dedicated restaurant menu system designed for consumer use. This work addresses that gap by asking: *How do embodied, multimodal interactions in a VR environment shape user perception, engagement, and decision confidence during the food selection process?*

3 SYSTEM DESIGN

3.1 Prototype Implementation

Figure 1: VR Restaurant Menus Prototype Interface

The Input^2 menu was developed in Unity (version 2022.3 LTS) targeting the Meta Quest 3S headset. The XR Interaction Toolkit was used to manage headset tracking, controller abstraction, and hand tracking input, allowing the system to recognize and respond to bare-hand gestures without physical controllers. The prototype renders a fully realized indoor restaurant environment, including tables, ambient lighting, decorative scene elements, and spatial audio comprising background restaurant ambience and music, all designed to support a sense of presence and ecological validity.

3.2 Menu and Food Models

The menu system presents users with four interactive 3D food items: a hot dog, a slice of pizza, a cup of coffee, and a can of Coca-Cola. Each item was modeled and textured to approximate a realistic visual representation of the dish. Food models are positioned on a virtual menu surface within the scene and can be grabbed, lifted, rotated, and inspected freely in three-dimensional space using hand tracking. This manipulation capability was central to our design goal of enabling users to evaluate food presentation from multiple angles, mimicking the visual inspection that occurs in real dining contexts.

3.3 Cart System

Once a user decides on an item, they can add it to a persistent virtual cart displayed within the environment, which tracks and summarizes their selections throughout the session.

4 METHODS

4.1 Participants

Eight participants were recruited through convenience sampling from the authors' university social and academic networks, including classmates and peers. All participants were university students. Prior VR experience varied across the sample: some participants had used VR headsets before in gaming or academic contexts, while others had no prior experience with immersive headsets. No formal exclusion criteria were applied beyond willingness to participate and absence of self-reported motion sickness that would prevent headset use. Participation was voluntary and unpaid.

4.2 Study Procedure

The study was conducted with participants completing the session in person. Participants used the Meta Quest 3S headset directly in a seated or standing position with the researcher present to assist with headset fitting and answer procedural questions. Each session lasted approximately 15 minutes or fewer, encompassing an orientation to the system, a free exploration period with the VR menu, and a post-task interview. No formal task script was enforced during the exploration phase; participants were encouraged to interact with the menu naturally and select items as they would in a real restaurant setting.

4.3 Data Collection

Following the hands-on session, each participant completed a semi-structured interview consisting of ten questions spanning both Likert-scale ratings and open-ended responses. The questions were designed to capture usability perceptions, food visualization quality, physical comfort, and likelihood of real-world adoption:

(1) How intuitive did you find the process of selecting and interacting with the food options in the virtual menu? *(1 = Very difficult, 5 = Very easy)*
(2) Did interacting with the 3D food models help you better understand how the dish looks and is presented?
(3) How would you rate the ease of navigation through the VR menu (e.g., grabbing, rotating, and exploring the food options)?
(4) Did the ability to rotate and manipulate the food models enhance your confidence in your food selection?
(5) Do you feel that the virtual representation of the food improved your ability to visualize the dish compared to traditional menus?
(6) How helpful did you find the VR menu in making your food choices?
(7) Was there any part of the VR interaction that you found difficult to use?
(8) Did you experience any discomfort, such as eye strain, dizziness, or motion sickness, while using the VR headset?
(9) How likely are you to use a VR menu like this in an actual restaurant setting?
(10) Would you recommend the use of a VR menu to others based on your experience?

Interviews were conducted verbally and responses were recorded through researcher notes taken during each session. No audio or video recording was used.

4.4 Analysis

We applied qualitative thematic analysis following the six-phase framework described by Braun and Clarke [4]: familiarization with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report. Both authors independently reviewed the interview notes and generated codes, then met to reconcile codes and agree on final themes. Likert-scale responses were summarized descriptively to support interpretation of the qualitative themes rather than as a basis for statistical inference, given the small sample size.

5 RESULTS

Thematic analysis surfaced findings across three primary dimensions corresponding to the core dimensions of our research question: usability and interaction, food visualization, and comfort and adoption.

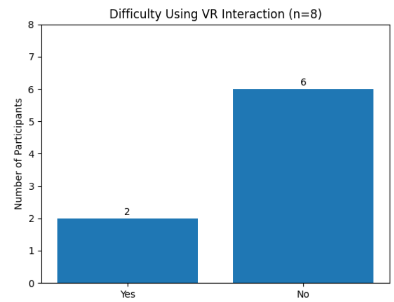

Figure 2: Difficulty Using VR Interaction

Participants generally found hand-tracked grabbing and rotation intuitive after a short acclimation period. Most described the grab mechanic as natural and satisfying once they understood the gesture required to initiate a hold. However, several participants noted a learning curve for fine-grained manipulation, particularly when attempting to orient a food item to a specific angle or place it back onto the menu surface precisely.

5.1 Food Visualization

The majority of participants reported that interacting with 3D food models improved their ability to evaluate dish presentation compared to traditional two-dimensional menus. Participants consistently described the ability to rotate and inspect items as a meaningful enhancement to their understanding of portion size, plating style, and overall dish appearance. Several noted that the spatial scale of food models experienced within the immersive environment, rather than on a flat screen, gave them a more accurate sense of what to expect from an actual order. This finding aligns with prior work on presence and behavioral realism in VR [16], and suggests that the environmental context of the restaurant scene contributed to the perceived authenticity of the food representations.

5.2 Comfort and Adoption

A subset of participants reported mild discomfort during the session, including eye strain and, in one case, mild motion sensitivity. These reports were more common among participants with no prior VR experience, suggesting an acclimation effect that might diminish with repeated use. Despite these concerns, the majority of participants expressed willingness to use a VR menu system in a real restaurant context, contingent on improvements to session setup time and food model fidelity. Participants with prior VR experience

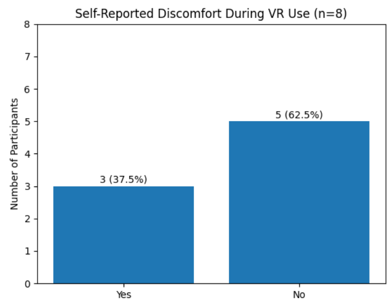

Figure 3: Self-Reported Discomfort During VR Use

were notably more enthusiastic about adoption, indicating that familiarity with the medium is a significant factor in acceptance.

6 DISCUSSION

Our findings suggest that embodied, hand-tracked VR menus can meaningfully improve food visualization and decision confidence relative to static alternatives. The ability to inspect 3D food models from multiple angles addresses a fundamental limitation of traditional menus and photograph-based digital interfaces, where a single fixed view may not convey portion size, texture, or plating accurately.

Recent analyses of consumer VR ecosystems further contextualize these findings. Steed et al.[11] argue that contemporary VR games have become a major driver of innovation in 3D user interface design, normalizing embodied, gesture-based, and multimodal interaction patterns. These design conventions increasingly shape user expectations for intuitive, controller-free interaction. Our results align with this trajectory: participants responded positively to hand-tracked manipulation, suggesting that VR menu systems may be most effective when they adopt interaction paradigms already familiar from consumer VR experiences.

Several limitations of this work should be acknowledged. The sample of eight participants, all university students recruited through convenience sampling, limits the generalization of our findings to broader consumer populations. Food model fidelity, which participants identified as an area for improvement, remains a significant design challenge for real-world deployment, where the visual gap between a virtual representation and an actual dish could affect trust and satisfaction.

Future work should conduct a controlled between-subjects study comparing the VR menu directly against a traditional photographic menu on standardized measures of decision confidence, satisfaction, and ordering accuracy. Improvements to food model realism, including texture detail and accurate scaling, would strengthen the ecological validity of the system. The inclusion of haptic feedback responses to grab events may further enhance the sense of physical presence during food inspection.

7 CONCLUSION

The results of this study indicate that embodied VR menus offer meaningful advantages over static two-dimensional alternatives in terms of food visualization quality and user decision confidence, while surfacing important design challenges around physical comfort, interaction learnability, and food model fidelity. These findings contribute empirical grounding to the emerging space of immersive hospitality interfaces and offer practical design implications for researchers and practitioners working at the intersection of extended reality and consumer experience.

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